



Sir Syed University
of Engineering & Technology

Computer Network Lab

Project Proposal

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Computer Network Lab Project

BS Computer Science, Batch 2022F

Company Network Design & Implementation Using Cisco Packet Tracer

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Project Proposal

1. Introduction:

In today's fast-paced business environment, secure, efficient, and scalable network infrastructure is crucial for companies to maintain productivity and seamless communication. This project focuses on designing and implementing a comprehensive enterprise network for a hypothetical company using Cisco Packet Tracer. The project will serve as a hands-on learning experience, showcasing skills in configuring, managing, and securing a network infrastructure in a simulated environment.

2. Project Overview:

The goal of this project is to design a company-wide network that meets the organization's operational needs, following best practices in hierarchical network design. The proposed network will be scalable, secure, and organized to optimize data flow and resource utilization. It will include various networking components and configurations, such as VLANs for segmentation, DHCP for dynamic IP address allocation, SSH for secure remote access, and NAT for enabling internet access. The network will be designed in Cisco Packet Tracer, which provides an environment for testing and troubleshooting network configurations before actual deployment.

3. Features:

- **Hierarchical Network Design:** Ensures modularity, scalability, and ease of management by implementing a structured approach to network layers (core, distribution, access).
- **VLANs and Inter-VLAN Routing:** VLAN segmentation to isolate different departments and Inter-VLAN routing via Switch Virtual Interface (SVI) for inter-departmental communication.
- **DHCP Server:** Automates IP addressing, enhancing scalability and reducing administrative overhead.
- **Port Security:** Configures port security to prevent unauthorized access and network breaches.
- **SSH (Secure Shell):** Provides secure, encrypted remote access to network devices.
- **NAT Overload (PAT):** Enables multiple internal hosts to share a single public IP for internet access.
- **Access Control Lists (ACLs):** Enhances security by controlling network traffic and restricting access to sensitive areas.
- **Wireless LAN (WLAN):** Configures access points to support secure and reliable wireless connectivity.
- **Static IPv4 Addressing and Host Configurations:** Sets up IP addresses and host settings for network consistency and easy management.

4. Technologies Used:

- **Cisco Packet Tracer:** Simulation tool for network design, configuration, and testing.
- **Hierarchical Network Design Model:** Core, Distribution, and Access layers for structured network layout.
- **VLANs and SVI (Switch Virtual Interface):** VLAN segmentation with Inter-VLAN routing.
- **Dynamic Host Configuration Protocol (DHCP):** For automatic IP allocation.
- **Port Security and Access Control Lists (ACLs):** Security configurations to protect data integrity and restrict access.
- **Secure Shell (SSH):** For encrypted and secure remote access.
- **Network Address Translation (NAT) and Port Address Translation (PAT):** For IP address management and internet connectivity.
- **Wireless LAN Configuration:** Setting up access points for wireless communication.

5. Implementation Plan:

1. Network Topology Design

- Design a hierarchical topology with core, distribution, and access layers.
- Define VLANs for each department (e.g., HR, Sales, Finance) and configure Inter-VLAN routing using SVI.

2. Device Configuration

- Configure routers, switches, and access points in Cisco Packet Tracer.
- Set up static IPv4 addresses and subnetting as per network requirements.

3. VLAN and Inter-VLAN Configuration

- Create VLANs for departmental segmentation and configure inter-VLAN routing on Layer 3 switches.

4. DHCP and Port Security Setup

- Configure DHCP servers to automate IP address assignment.
- Implement port security to restrict unauthorized device connections.

5. Security Measures

- Implement SSH for secure remote access to network devices.
- Configure Access Control Lists (ACLs) to control access to sensitive data.

6. Internet Access Configuration

- Set up NAT Overload (PAT) to enable multiple internal users to share a single external IP.
- Test internet connectivity and resolve any routing or address translation issues.

7. Wireless Network Setup

- Configure and secure wireless access points to provide Wi-Fi connectivity.

8. Testing and Troubleshooting

- Test each segment and connection in Cisco Packet Tracer.
- Troubleshoot and resolve any connectivity, security, or configuration issues.

6. Expected Outcome:

Upon completion, the network will allow secure and efficient communication across different departments within the company. Users will experience seamless connectivity within their VLANs, and inter-departmental communication will be possible while maintaining security and access control. The network will be optimized for resource utilization, allowing the company to scale its network infrastructure in the future. Wireless connectivity will further enhance accessibility for mobile and IoT devices within the organization.

7. Conclusion:

This project provides a realistic simulation of designing, implementing, and managing a complex enterprise network. It highlights the importance of network segmentation, security, scalability, and structured design in a corporate environment. By following best practices in network design and employing Cisco technologies, this project will equip students with valuable skills applicable to real-world networking scenarios. This proposal offers a clear plan and framework to build an efficient, secure, and scalable company network using Cisco Packet Tracer.